

Indian Statistical Institute
Semester 1 (2024-2025)
M. Math. 1st Year
Midterm
Measure Theory

Date: 13.9.24

Time: 1 hour 30 minutes

Total Points: 30

Answers must be justified with clear and precise arguments. More than one uncrossed answers to the same question or part of a question will not be entertained and only the first uncrossed answer will be graded. If you use any theorem or result proved in class state it explicitly.

1. The sum of two sets $A + B$ is defined as $A + B = \{x + y : x \in A, y \in B\}$ (which also equals $\cup_{x \in A} \{x + B\}$) where A, B are subsets of $\mathbb{R}$.

(a) Show that if A is countable and B is measurable then $A + B$ is measurable.

(b) What can you say about the measurability of $A + B$ if A is open and B is measurable? 3+3 = 6 pts.

2. Suppose f defined on $\mathbb{R}$ is nonnegative such that $\int_{\mathbb{R}} f^p dm < \infty$, where $1 \leq p < \infty$. Show that given $\epsilon > 0$ one can find $\delta > 0$ such that for any measurable subset A , $m(A) < \delta$ implies $\int_A f^p dm < \epsilon$. (You may start by writing $f = f1_{\{0 \leq f \leq M\}} + f1_{\{f > M\}}$ and choosing M first.) 6 pts.

3. A result of Euler says that for $\alpha > 0$, as $n \uparrow \infty$,

$$\Gamma(\alpha, n) = \frac{n!n^\alpha}{\alpha(\alpha+1)(\alpha+2) \cdots (\alpha+n)} \rightarrow \Gamma(\alpha) = \int_0^\infty e^{-t} t^{\alpha-1} dt,$$

(also expressed as $(1+\alpha)(1+\frac{\alpha}{2}) \cdots (1+\frac{\alpha}{n}) \sim \frac{n^\alpha}{\Gamma(\alpha+1)}$).

(a) Show that $\Gamma(\alpha, n) = n^\alpha \int_0^1 (1-s)^n s^{\alpha-1} ds = \int_0^n (1-\frac{t}{n})^n t^{\alpha-1} dt$, where the first equality is the main one to be proved, the other following from a change of variable.

(b) Justify the limit in Euler's result. 3 + 3 = 6 pts.

4. Let f_n be a sequence of functions in L^p , $1 \leq p < \infty$, which converge a.e. to a function $f \in L^p$. Show that if f_n converges to f in L^p then $\|f_n\|_p \rightarrow \|f\|_p$. 6 pts.

5. (a) A result of Lebesgue says that a bounded function f (on $[0, 1]$ say) is Riemann integrable iff its set of discontinuity points has measure zero. Consider the Dirichlet function $f(x) = 0$, if $x \in Q \cap [0, 1]$, 1, if $x \in Q^c \cap [0, 1]$. Find the set of discontinuity points of this function.

(b) Let Ω be a countably infinite set and let $\mathcal{F}$ be the collection of subsets of Ω which are either finite subsets of Ω or complements of finite subsets of Ω . Is $\mathcal{F}$ an algebra of sets? 3 + 3 = 6 pts.